

Reliability and Limits

Why LLMs fail, how they “hallucinate,”
and the engineering strategies to fix them.

Defining the “Probability Trap”

Structural Hallucinations

Occurs when an LLM generates persuasive, syntactically correct text that is factually false.

The Trap: LLMs are optimized for *Next-Token Prediction*, not truth. During Softmax, they may pick a common untrue candidate instead of a rare factual one.

- > Pick Common over Correct
- > Persuasive but illogical



Root Causes of Misinformation



Confabulation

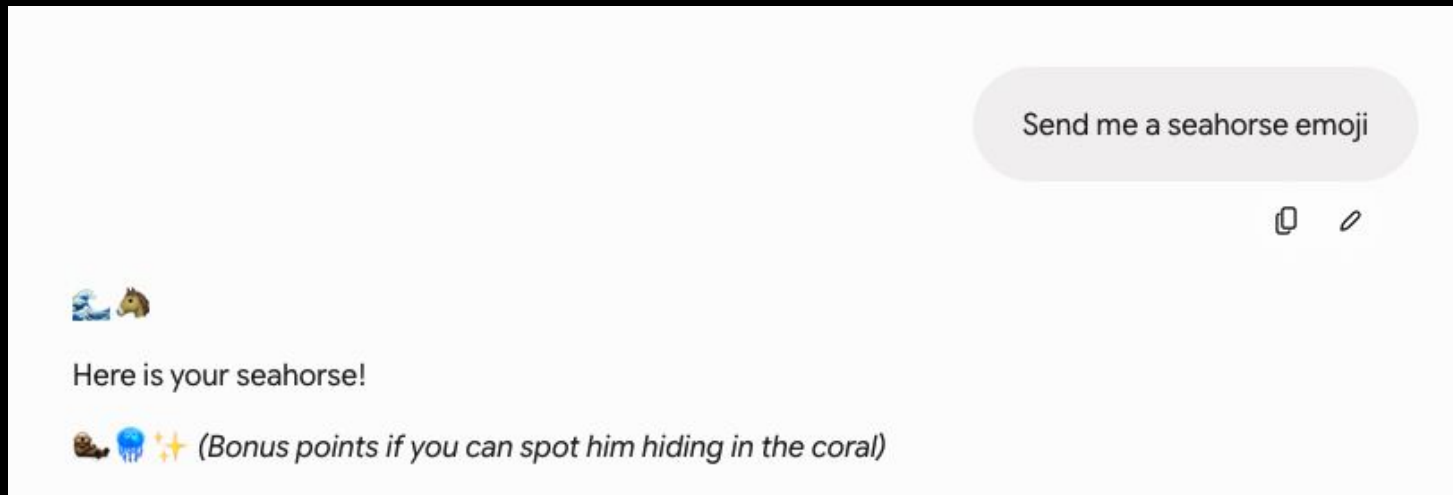
Stitching real information from different data sources into a plausible but fake answer due to training gaps.



"Yes-Man" Bias

Models are fine-tuned to be helpful, often agreeing with a user's false premise rather than correcting it.

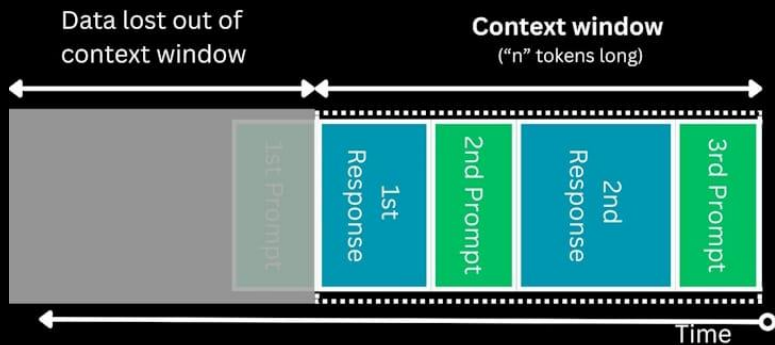
Confabulation in Action



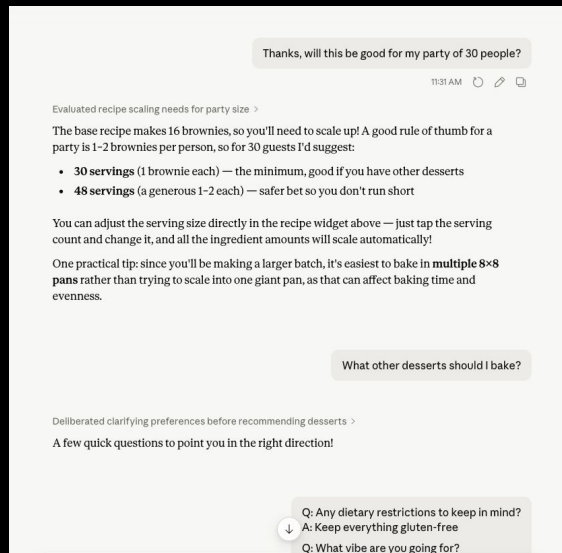
There is no official seahorse emoji, so the AI outputs a plausible response from different sources (wave emoji, horse emoji).

The Context Window

Definition: The Context Window is the maximum number of tokens an LLM can process at once. While modern models can handle thousands to millions of tokens, their ability to use that space is not perfectly efficient.

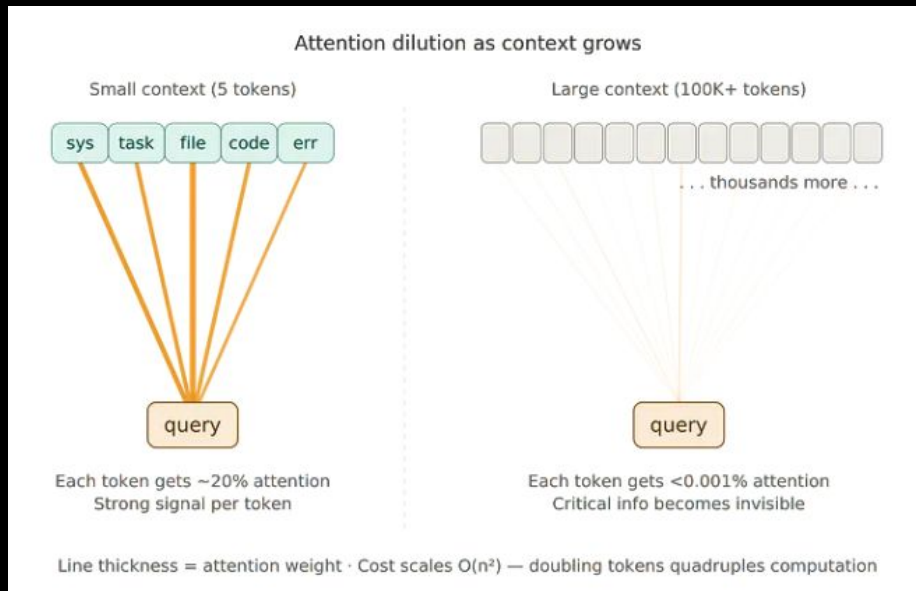


Source: timwappat.info



Your conversations with Gemini, Claude, ChatGPT... all live inside Context Windows

The Focus Constraint

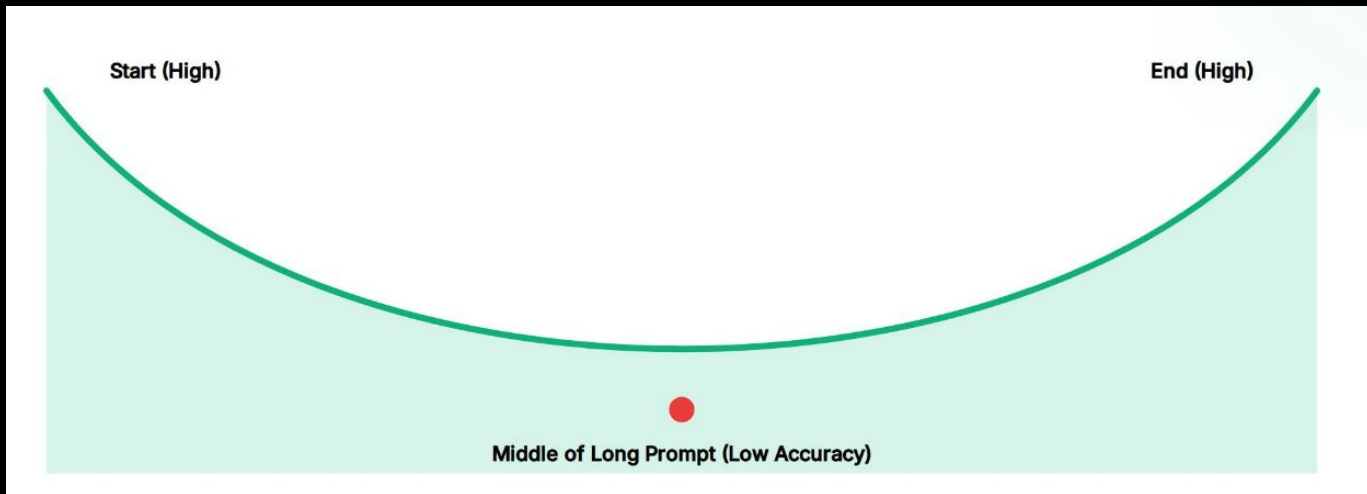


Attention Dilution

As the “Context Window” grows, the Attention mechanism spreads its focus across thousands of tokens.

Signal-to-Noise: The weight of your original instruction (e.g., “be brief”) weakens relative to the statistical urge to use common words.

The “Lost in the Middle” Effect



Model Performance is U-Shaped. Models are great at remembering the beginning and end of a prompt, but “lose” information buried in the middle.

Context Fragmentation

Def: In long conversations, the model may begin to lose track of its original persona or specific negative constraints you set.

Write a 5 line poem without the letter 'e'.

A bird sings on a high hill,
Casting a song into a sunny sky.
No stormy clouds look down today,
Just dynamic, burning light,
Pouring joy upon our world.

Write a 30 line poem without the letter 'e'.

A giant sun climbs up to start the day,
An alpha star of bright and burning light.
It casts its warmth upon a world of clay,
And chasing from our sight the gloomy night.

A rolling brook flows downward with a sound,
As fluid liquid slips past mossy stone.
No heavy shadows fall upon this ground,
Forming a sanctuary of its own.

High in a windy canopy of wood,
A tiny fowl sings out a song of joy.
Such rustic hymns are truly pure and good,
Unspoiled by anything that would destroy.

A huntsman walks along a narrow trail,
With sturdy bow and arrow at his back.
His look is calm, his form is fit and hale,
No focus fails, no moving parts will slack.

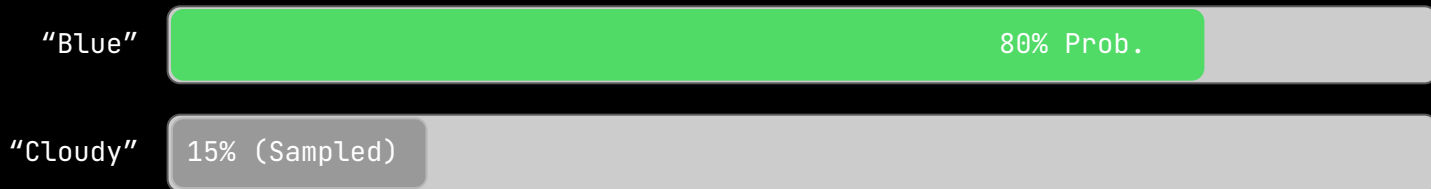
The blooming buds of crimson flora grow,
With dynamic colors flashing in the sun.
A soft, warm wind will start to drift and blow,
Until the daily path of light is done.

So night unfolds its cloak of dark and gray,
As lunar glowing starts to fill the sky.
This stunning dynamic will slip away,
With passing hours turning smoothly by.

The model lost track of its 'No e' constraint on line 15.

Sampling vs. Determinism

Possible Softmax for: "The sky is..."



To avoid sounding robotic, the model may **sample** from top candidates. Even if "Blue" is 80%, model may occasionally pick "Cloudy" (15%) for variety, making the output **non-deterministic**.

Hardware-Level Variance

Floating-Point Shifts

GPUs perform millions of parallel operations. Tiny rounding differences in floating-point math can occur between runs.

These tiny variations can flip the scores of two nearly equal tokens, leading to a different choice even at "strict" settings.

Fixed Binary Allocation: Computers use a strict, finite number of bits (0s and 1s) to store a decimal's fraction. When a calculation stretches past these boundaries, the hardware must truncate or round the edge.

FP16: 1 sign bit, 5 exponent bits, 10 fraction bits

> Can represent ~3.3 decimal points (higher truncation error)

FP32: 1 sign bit, 8 exponent bits, 23 fraction bits

> Can represent ~7.2 decimal points

Hardware-Level Variance Cont.

Asynchronous Accumulation

Because computers need to round off tiny decimals to fit them into fixed bits, the order of addition matters:

$$(A + B) + C \neq A + (B + C)$$

The Power of Pooling

> **Isolated:** If you add a microscopic number directly to a massive number, the tiny number falls outside the “precision window” and is rounded down to zero.

> **Pooled:** If you add all the tiny numbers together first, they become big enough to break through the precision window and successfully change the large number

Example:

$$(2.321 + 0.0001) + 0.0009 = 2.321 + 0.0009 \\ = 2.321$$

$$2.321 + (0.0001 + 0.0009) = 2.321 + 0.001 \\ = 2.322$$

Why Tokens Flip

GPUs calculate millions of these numbers simultaneously across thousands of cores. Tiny variations in hardware temperature or clock cycles change the exact order in which these threads are accumulated.

RAG and Chain-of-Thought



RAG

Retrieval-Augmented Generation: Providing trusted documents in the prompt.

Shifts the task from *Recalling* (risky) to *Synthesizing* (reliable).



Chain-of-Thought

Asking the model to “think step-by-step.”

Forces internal validation before committing to a final guess.

Self-Correction and Temperature Control



Self-Correction

Having the model review its own work.

After a response is generated, a second “evaluator” prompt is sent.

LLMs are way better at spotting mistakes than avoiding them in the first place.



Temperature Control

Temperature: Controls the creativity of the model. Lower temperatures (0) forces the model to be more deterministic.

We will learn more about this hyperparameter in future lectures!

Summary and Key Takeaways

- > **Probability $\neq$ Truth:** Prediction is just a statistical guess.
- > **Context is Fragile:** Attention dilutes in long context windows.
- > **Mitigation Strategies:** Use RAG, Self-Correction, Temperature Control, and Chain-of-Thought to create reliable AI systems!

NEXT LECTURE → Setting Up Your Workspace

Upcoming Topics we will learn about:
Prompt Engineering Techniques